

One-Period Binomial Option Pricing

Live-Data Implementation & Streamlit Dashboard

ES 418 — Financial Modelling and Engineering

Ayush Kumar Vansh Goel Yug Mitulkumar Desai Mohit Kumar Yadav

Group 4 — Instructor: Prof. Mithun Radhakrishna

Indian Institute of Technology Gandhinagar

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Presentation Outline

- 1 Introduction & Motivation
- 2 Mathematical Framework
- 3 Case Study: META Platforms
- 4 Streamlit Dashboard
- 5 Limitations & Conclusion

What Are Options?

Definition

An **option** is a derivative contract giving the holder the *right, but not the obligation* to buy/sell an underlying asset at a pre-agreed **strike price** K on/before expiry.

- **Call option** — right to *buy*; payoff
 $= \max(S_T - K, 0)$
- **Put option** — right to *sell*; payoff
 $= \max(K - S_T, 0)$
- **European** options: exercisable only at expiry

Our Approach

- Cox–Ross–Rubinstein (1979)
one-period binomial model
- Live data from **Yahoo Finance**
- Full pipeline: volatility $\rightarrow$ tree $\rightarrow$ price $\rightarrow$ signal
- Interactive **Streamlit dashboard**

The Pricing Problem

How much should you pay *today* for this right?

Task 1 — Binomial Tree Construction

Up/Down Factors from Volatility

Given annualised volatility σ and maturity T (years):

$$u = e^{\sigma\sqrt{T}}, \quad d = e^{-\sigma\sqrt{T}} = \frac{1}{u}$$

End-of-period prices:

$$S_u = S_0 u, \quad S_d = S_0 d$$

No-Arbitrage Condition

$$d < e^{rT} < u$$

The risk-free growth must lie *strictly between* d and u .

Realized Volatility

Computed from 60-day rolling window of log returns:

$$\sigma_{\text{daily}} = \text{std}\left(\ln \frac{S_i}{S_{i-1}}\right)$$

$$\sigma_{\text{annual}} = \sigma_{\text{daily}} \times \sqrt{252}$$

META Example

$$S_0 = \$670.91, \quad \sigma = 43.12\%$$

$$T = 0.04689 \text{ yr} \approx 17 \text{ days}$$

$$u = 1.0979, \quad d = 0.9108$$

$$S_u = \$736.58, \quad S_d = \$611.09$$

Task 2 — Option Payoffs

European Call ($K = \$670$)

$$C_u = \max(S_u - K, 0) = \max(736.58 - 670, 0)$$

$$C_d = \max(S_d - K, 0) = \max(611.09 - 670, 0)$$

European Put ($K = \$670$)

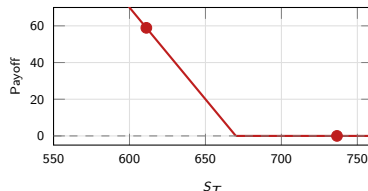
$$P_u = \max(K - S_u, 0) = \max(670 - 736.58, 0) = 0$$

$$P_d = \max(K - S_d, 0) = \max(670 - 611.09, 0) = \$58.91$$

Interpretation

- If META rises to \$736.58, the put **expires worthless** — you wouldn't sell at \$670 when market is at \$736.
- If META falls to \$611.09, the put **pays \$58.91** — you can sell at \$670 when market is only \$611.

Put Payoff Profile



Task 3 — Replicating Portfolio

Two-Equation System

Portfolio of Δ shares + B in risk-free bond replicates option:

$$\Delta \cdot S_u + B e^{rT} = C_u \quad (\text{up state})$$

$$\Delta \cdot S_d + B e^{rT} = C_d \quad (\text{down state})$$

Solution

$$\Delta = \frac{C_u - C_d}{S_u - S_d}$$

$$B = e^{-rT} (C_u - \Delta S_u)$$

Δ = option delta (hedge ratio)

B = bond position (PV of cash leg)

META Put Calculation

$$\Delta = \frac{0 - 58.91}{736.58 - 611.09} = \frac{-58.91}{125.49} = -\mathbf{0.4694}$$

$$B = e^{-0.05 \times 0.04689} (0 - (-0.4694) \times 736.58)$$

$$B = e^{-0.002345} \times 345.73 = \mathbf{\$344.95}$$

Portfolio Interpretation

Short 0.4694 shares of META

Long \$344.95 in risk-free bonds

This *exactly* replicates the put payoffs in both states.

Task 4 — Option Pricing: Two Methods

Method 1: Replication

No-arbitrage: the option price equals the cost of replicating portfolio:

Formula: $V_0 = \Delta \cdot S_0 + B$

$$V_0 = (-0.4694)(670.91) + 344.95$$

$$V_0 = -314.94 + 344.95 = \text{\$30.017}$$

Method 2: Risk-Neutral Pricing

Risk-neutral probability p^* :

$$p^* = \frac{e^{rT} - d}{u - d} = \frac{1.002346 - 0.91085}{0.18703} = 0.4892$$

$$V_0 = e^{-rT}[p^* C_u + (1 - p^*) C_d] = \text{\$30.017} \checkmark$$

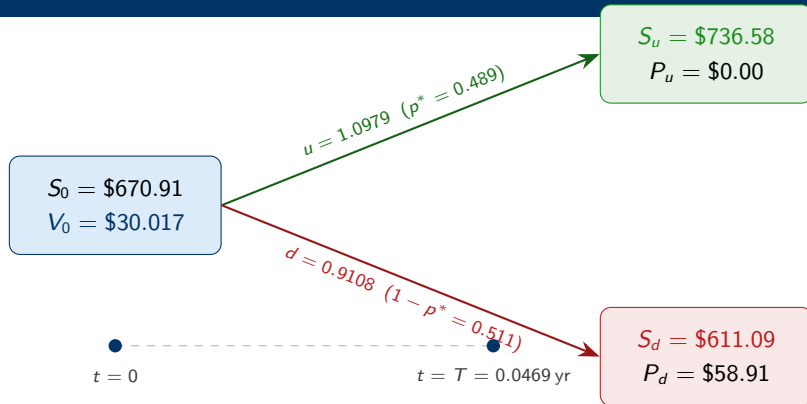
Consistency Verification

Both methods agree to machine precision ($< 10^{-8}$), confirming no-arbitrage consistency.

Risk-Neutral Intuition

- $p^* = 0.4892$: the *risk-neutral* probability of an up move — not a real-world probability
- Under p^* , the expected stock return equals the risk-free rate
- We discount at r , not at the stock's expected return
- No assumptions needed about investor risk preferences

One-Period Binomial Tree — META Put



Replicating Portfolio: $\Delta = -0.4694$ shares, $B = \$344.95$ bond

Strike $K = \$670$ — Expiry: 2026-05-08 — European Put

Investment Signal & Key Results

Key Results Summary (20 April 2026)

Param	Value	Param	Value
S_0	\$670.91	u, d	1.0979, 0.9108
K	\$670.00	S_u, S_d	\$736.58, \$611.09
T	0.04689 yr	P_u, P_d	\$0.00, \$58.91
r	5.00%	p^*	0.4892
σ	43.12%	Δ	-0.4694
Model Price		\$30.017	
Market Price		\$27.125	

Investment Signal

$$\begin{aligned}\text{Mispricing} &= \frac{V_0^{\text{model}} - V_0^{\text{market}}}{V_0^{\text{market}}} \\ &= \frac{30.017 - 27.125}{27.125} = +10.66\%\end{aligned}$$

BUY — Model > Market
Option potentially undervalued

10.66% > 5% threshold $\Rightarrow$ **BUY**

Streamlit Dashboard — Overview

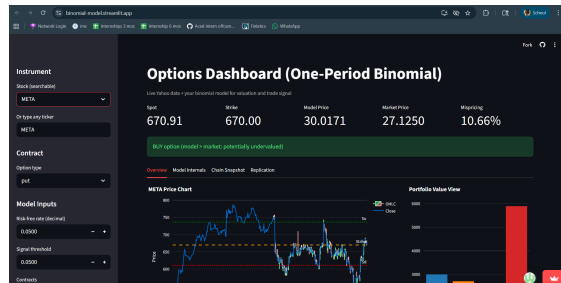
❑ Live Web Application

`binomial-model.streamlit.app`

- Fetches **live Yahoo Finance** data on every run
- Works for *any* ticker in the option chain
- All four assignment tasks computed automatically

Left Sidebar Controls

- Stock (searchable) & Option type (call/put)
- Risk-free rate & Signal threshold
- Volatility: daily/intraday & rolling window
- Option expiry & strike (auto ATM)
- Horizon: full expiry vs. 1-hour step



Dashboard — Header KPIs & Overview Tab

Header KPI Cards

Five real-time metric cards are always visible:

Spot	Strike	Model	Market	Misprice
\$670.91	\$670	\$30.02	\$27.13	+10.66%

A colour-coded signal banner (green/red/grey) immediately shows the trading decision.

Tab: Overview

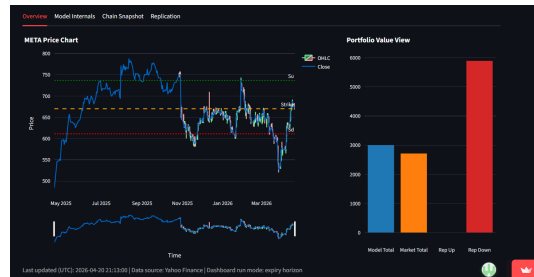
- **Price Chart** — 1-year OHLC candlestick with S_u , S_d , strike lines
- **Portfolio Value View** — grouped bar chart: Model Total, Market Total, Replicating Up/Down values

Options Dashboard (One-Period Binomial)

Live Yahoo data + your binomial model for valuation and trade signal

Spot	Strike	Model Price	Market Price	Mispricing
670.91	670.00	30.0171	27.1250	10.66%

BUY option (model > market: potentially undervalued)



Dashboard — Model Internals, Chain Snapshot & Replication

Tab: Model Internals

All intermediate variables in monospace green font: σ , T , r , u , d , S_u , S_d , payoffs, p^* , Δ , B , model price, market price, threshold.

Tab: Chain Snapshot

Full option chain from Yahoo Finance for chosen expiry: strike, bid, ask, volume, open interest, implied volatility, ITM flag. Selected strike row highlighted.

Tab: Replication

Scaled to number of contracts (default: 100):

- Shares qty = $\Delta \times 100 = -46.94$
- Cash today = $B \times 100 = \$34,495$
- Portfolio value: up ($\approx \$0$) & down ($\$5,890$)

Overview **Model Internals** Chain Snapshot Replication

- Symbol: META
- Option: PUT
- Expiry: 2026-05-08
- Vol source: daily close returns, window=60
- Sigma (annualized): 43.1234%
- Maturity T: 0.04680365 years
- Risk free r: 5.4669%
- u, d : 1.497882, 0.936845
- S_u, S_d : 190.5099, 811.8948
- Payoff Up/Down: 0.0000 / 58.9892
- Risk-neutral p: 0.499222
- Delta: -0.469429
- Bond B: 344.355342
- Model option price: 30.0171
- Market option price: 37.1258
- Decision threshold: 0.000

Last updated (UTC): 2026-04-20 21:13:00 | Data source: Yahoo Finance | Dashboard run mode: expiry horizon

Overview **Model Internals** Chain Snapshot Replication

	strike	bid	ask	volume	openinterest	impliedvolatility	itmflag
0	120	0.05	0	0.04	25	1	2.2889
1	140	0.04	0	0.04	27	14	2.1875
2	160	0.05	0	0.04	46	21	2.0891
3	180	0.12	0	0.04	38	8	1
4	190	0.30	0	0.04	25	47	1.6219
5	195	0.46	0	0.04	100	100	1.6125
6	198	0.51	0	0.1	3	120	1.6204
7	199	0.52	0	1.88	46	95	1.4988
8	200	0.50	0	0.1	26	24	1.6053
9	201	0.53	0.01	0.01	220	98	1.7109

Selected strike row:

	strike	bid	ask	volume	openinterest	impliedvolatility	itmflag
5	195	0.46	0	0.04	100	100	1.6125

Overview **Model Internals** Chain Snapshot Replication

Contracts x Size	Replicating Shares Qty	Replicating Cash Today	Replicating Value Now
100	-46.9420	34495.5342	3001.7061

- Replicating portfolio in Up state: -0.0000
- Replicating portfolio in Down state: 5890.5246
- Cash grown to step: 34576.5893
- Model total value: 3891.7963
- Market total value: 2712.5898

Last updated (UTC): 2026-04-20 21:13:00 | Data source: Yahoo Finance | Dashboard run mode: expiry horizon

Limitations & Future Extensions

Current Limitations

- **Single period:** no path-dependence
- **Realized vs. implied volatility:** historical $\sigma \neq$ implied σ (vol smile/skew ignored)
- **No dividends:** discrete dividends reduce stock price at ex-date
- **European only:** American options need backward induction
- **Frictionless markets:** no bid-ask spreads or margin costs

Natural Extensions

- Multi-period binomial tree
- Black-Scholes as continuous limit ($n \rightarrow \infty$)
- Greeks dashboard: Γ , Θ , Vega
- Implied volatility surface
- American option pricing (backward induction)
- Dividend-adjusted model

Conclusion

What We Achieved

- **Full end-to-end pipeline:** live data $\rightarrow$ volatility $\rightarrow$ tree $\rightarrow$ price $\rightarrow$ signal
- Both pricing methods (*replication* and *risk-neutral*) verified to agree to machine precision
- Interactive Streamlit dashboard accessible for any ticker

META Case Study Summary

Item	Value	Note
Model price	\$30.017	Replication & risk-neutral Bid-ask midpoint
Market price	\$27.125	
Mispricing	+10.66%	> 5% threshold
Signal	BUY	Potentially undervalued

→ **Live App:** binomial-model.streamlit.app

Thank You!

Group 4

Ayush Kumar

Python / Colab

Vansh Goel

Streamlit Dashboard

Yug Mitulkumar Desai

Report Writing

Mohit Kumar Yadav

Presentation

→ `binomial-model.streamlit.app`